

Tasks Module, Part 4: The Ropeway Analysis

Which workers could take the jobs the ropeway may bring, and how many may be left without work

Remember. The workers are **made-up**. The number of ropeway jobs is a **placeholder** (we do not know it yet). The job task lists are **provisional**, taken from the national occupation list. The results show how the method works and what it can and cannot say. They are not a forecast.

1. The question

If the ropeway takes work away from people who carry pilgrims and goods on the trek, how many of them might end up without work? Two things decide this. First, *how many new jobs there are*. Second, *whether the people who lose work can actually do those jobs*. The task questions help with the second thing.

2. Which jobs might people move to?

We listed 13 jobs that a ropeway and the trade around it may bring. Each has a short list of tasks it needs. The lists come from the national occupation list (NCO 2015), so we can say where each one comes from. Where the list does not cover a job, we used judgement and say so. Some jobs also need a paper (a licence or certificate). We call that a **gate**: without it the person cannot take the job, however many tasks they know.

Id	Job	Tasks it needs	Where from	Gate
RW1	Aerial ropeway operator	8 operate an engine or machine; 12 check equipment for safety; 24 coordinate by phone, radio, signals	NCO 8343.1700	none
RW2	Electrician / wireman (ropeway plant)	10 electrical or wiring work; 12 check equipment for safety	NCO 7411.0301 (QP ELE/Q7302, NSQF 3)	ITI or NSQF-3 electrical certificate
RW3	Mechanic / fitter (ropeway plant)	8 operate an engine or machine; 9 repair by hand	NCO 7231.0500 (nearest entry)	ITI or technical certificate (assumed)
RW4	Station loading and crowd attendant	3 load, unload, stack, count goods; 23 keep order or watch property; 24 coordinate by phone, radio, signals	NCO 9333 loader + 8343.1700 attendant; judgement	none
RW5	Ticket and information counter clerk	16 handle cash and payments; 25 keep records or accounts; 21 guide or explain to visitors	NCO 4224.0100 / 4221 (nearest); judgement	none
SEC	Security guard / gateman	23 keep order or watch property; 21 guide or explain to visitors; 25 keep records or accounts	NCO 5414 Gate-man/Unarmed Security Guard (QP SKS/Q0101, NSQF 4)	NSQF-4 guard certificate
HOS1	Room attendant / housekeeping	20 clean rooms or public areas	NCO 5151.0202	none
HOS2	Waiter / food and beverage steward	19 serve guests; 16 handle cash and payments	NCO 5131.0401 (NSQF 3, training-linked)	none
HOS3	Cook	18 cook or prepare food or drink	NCO 5120.0300 / 5122	none
TRA	Light motor vehicle (car) driver	7 drive a motor vehicle; 12 check equipment for safety; 16 handle cash and payments	NCO 8322.0100	Car (LMV) licence

Id	Job	Tasks it needs	Where from	Gate
SHP	Shop assistant	14 sell goods or services; 4 pack or tie goods	NCO 5223.0200	none
GDE	Tourist guide	21 guide or explain to visitors	NCO 5113.0200	Guide or trekking permit (assumed; not in NCO)
CON	Construction labourer (build phase)	11 build or construct; 1 carry loads on foot; 3 load, unload, stack, count goods	NCO 9313.9900	none

Two cautions. (1) The national list gives only 1 to 3 tasks for most of these jobs. That is few (see section 6). (2) The gate for ropeway operator, mechanic and guide is our assumption, since the national list does not state one. Before real use, these lists should be checked with the ropeway company, hotel owners and unions.

3. Two simple measures

For each worker and each job we count:

- **Shortage:** of the tasks the new job needs, what share does the worker *not* do? 0 means they can do all of them. 1 means none.
- **Unused:** of the tasks the worker does now, what share would the new job not use? High means a lot of their skill goes to waste.

(These two ideas come from Neffke and co-authors, 2024, but their version uses many skills per job; ours uses tasks that are answered yes or no.) We use two views of what a worker can do. **Strict:** only tasks done regularly now. **Broad:** tasks done regularly plus tasks done before (answer 2).

Rule fixed before looking at the data: a worker *could take* a job if the shortage is one-third or less, *and* the worker holds any gate paper. The base case uses the broad view. We also test other cut-offs (0 and one-half).

Worked example. Worker 106 is a palki. The job “construction labourer (build phase)” needs: 11 build or construct; 1 carry loads on foot; 3 load, unload, stack, count goods. Doing tasks regularly now, the worker lacks 2 of the 3 (shortage 0.67), so the strict view says *no*. The worker also did before: 3 load, unload, stack, count goods. In the broad view the shortage is 0.33 and the answer is *yes*. This is the reason we ask answer 2 on the task grid.

4. Who is exposed to the ropeway?

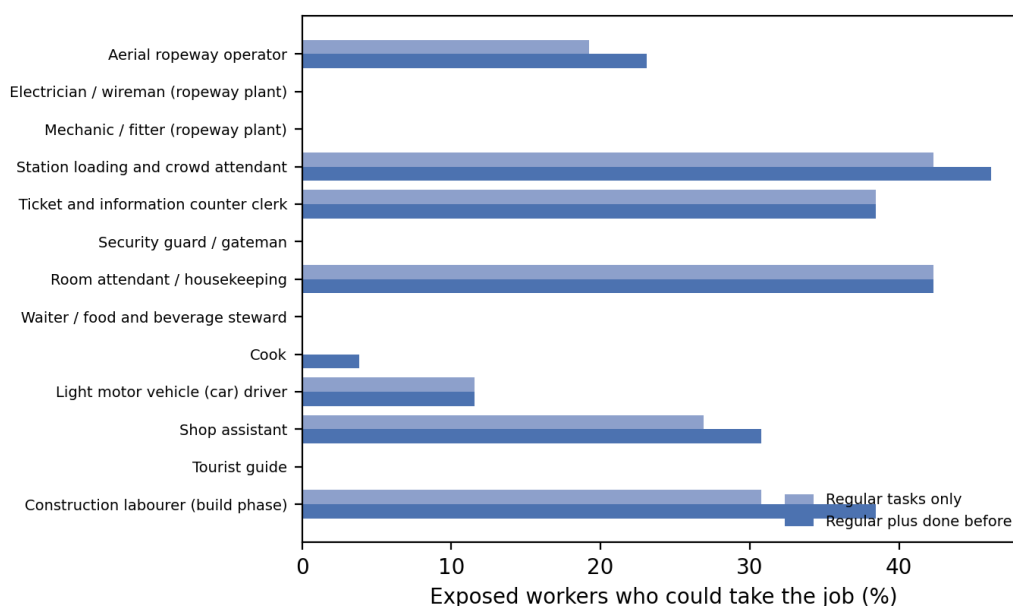
We need to know whose work the ropeway may replace. The test data gives three ways to decide:

Way of deciding	Workers
By job title: porters, palki bearers, pony workers, pony owners	26
By main task: carrying, passenger, or pack animals	23
By any regular carrying, passenger or pack-animal task	66

The first is used in the base case (26 workers). The second misses pony people whose main task is feeding animals, loading, or selling rides, and it includes one wage labourer. The third counts 66 workers, including many shop and hotel workers who sometimes carry loads. This is too wide. **Lesson.** Exposure should not be guessed from a task or a title. The survey needs its own direct exposure questions (module D), such as “does your income come from taking people or goods on the Kedarnath trek?”.

5. Who could take which job?

The picture and table show, for the 26 exposed workers, the share who pass the rule for each job.



Share of exposed workers who could take each job (shortage one-third or less, and gate paper held).

Job	Tasks	Hold gate, all workers (%)	Strict (%)	Broad (%)	Broad, no gate (%)
Aerial ropeway operator	3	100	19	23	23
Electrician / wireman (ropeway plant)	2	10	0	0	0
Mechanic / fitter (ropeway plant)	2	10	0	0	0
Station loading and crowd attendant	3	100	42	46	46
Ticket and information counter clerk	3	100	38	38	38
Security guard / gateman	3	3	0	0	0
Room attendant / housekeeping	1	100	42	42	42
Waiter / food and beverage steward	2	100	0	0	0
Cook	1	100	0	4	4
Light motor vehicle (car) driver	3	17	12	12	12
Shop assistant	2	100	27	31	31
Tourist guide	1	3	0	0	0
Construction labourer (build phase)	3	100	31	38	38

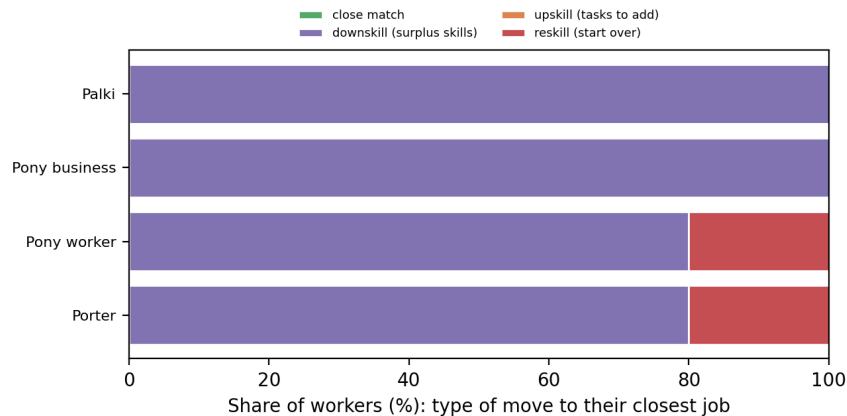
- The jobs closest to carrying work (station attendant, room attendant, construction, ticket counter) suit 38–46 in 100 exposed workers.
- The technical ropeway jobs (electrician, mechanic) suit almost no one. Few people hold an ITI or similar paper, and fewer do electrical or engine work.
- Guard, guide and driver jobs are open on tasks to many, but few hold the paper. **Gates matter more than tasks for these jobs.** This is why the questions on licences (T3) are in the survey.
- On average an exposed worker could take **2.3** of the 13 jobs. **15 in 100** could take none at all (19 in 100 in the strict view). Weighted for drop-out, 16 in 100.

The number who could take *no* job is important. It does not depend on how many jobs exist. It is a floor: even if jobs were unlimited, these workers could not be placed with the tasks and papers they hold now.

6. Type of move, and a warning

Neffke and co-authors sort moves into four types by combining shortage and unused skills: a *close match* (low, low), *downskilling* (few tasks to add, many unused), *upskilling* (many to add, few unused), and *reskilling* (many to add, many unused).

Type of move	All worker-job pairs (%)	To the worker's closest job (%)
close match	1	0
downskill (surplus skills)	27	88
upskill (tasks to add)	0	0
reskill (start over)	73	12



Type of move to each worker's closest job, by present job.

Almost every move is called “downskilling” or “reskilling”, and 99 in 100 pairs have a high “unused” share. This is not a finding. It happens because our job lists have only about 2.2 tasks while a worker does about 7.4, so most of what the worker does can never be used by a short list. **Lesson.** The “unused” measure and the four types need long task lists for each job (10 or more). With the

short lists we have, use **shortage only**. To use the four types, the survey needs job task lists from employers or a ropeway operator.

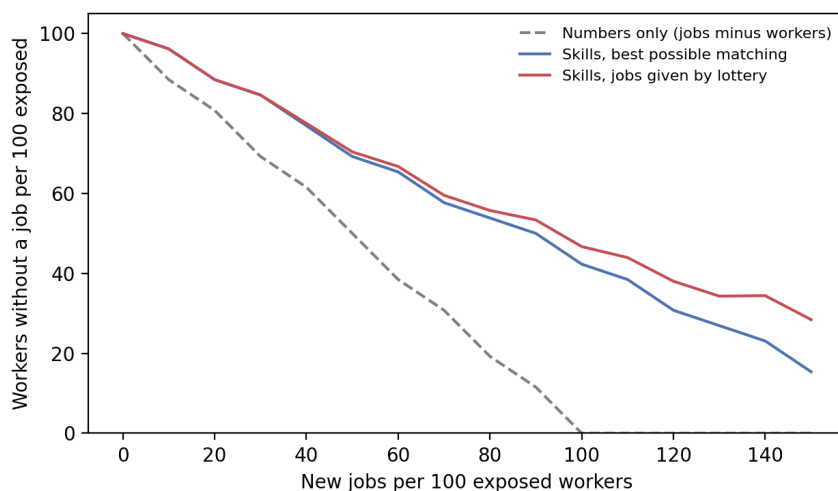
We also checked whether the closest job changes with the measure. The closest job by shortage is the same as the closest by two other common measures (cosine and Jaccard) for 62 in 100 workers. So the ranking of jobs *is* sensitive to the measure when job lists are short. Shortage is the only measure with a plain meaning (“how many of the new job’s tasks are new to me”), so we keep it.

7. How many may be left without work?

We count three things, all per 100 exposed workers:

- **Numbers only:** exposed workers minus new jobs. This is what most reports do. It ignores who can do what.
- **Skills, best possible:** give jobs to workers so that as many as possible are placed, each worker only in a job they could take (maximum matching). This is the best case.
- **Skills, by lottery:** jobs are given at random among workers who could take them, repeated 300 times. This is closer to how hiring might go.

Job supply is a placeholder. We tried three totals (25, 50 and 100 new jobs per 100 exposed workers) and three mixes (equal across job types; three times as many technical jobs; three times as many service jobs). The real numbers must come from the ropeway plan and an employer check.



Workers left without a job per 100 exposed, as new jobs increase (equal mix).

New jobs per 100 exposed	Mix	Numbers only	Skills, best	Skills, lottery
25	equal	77	88	88
25	technical-heavy	77	88	88
25	service-heavy	77	85	85
50	equal	50	69	70

New jobs per 100 exposed	Mix	Numbers only	Skills, best	Skills, lottery
50	technical-heavy	50	73	73
50	service-heavy	50	73	74
100	equal	0	42	47
100	technical-heavy	0	46	51
100	service-heavy	0	42	45

Reading the table. In the middle case (50 new jobs per 100 exposed, equal mix), numbers alone give 50 without a job per 100. Once tasks and papers are taken into account, it is **69** in the best case and **71** by lottery. Resampling the workers 1,000 times gives a range of about 69 to 73 for the best case. That range comes only from having few workers; it does not include uncertainty about the job supply or the job lists, which are larger.

Even if there were **one new job for every exposed worker**, about 42 in 100 would still be without work, because the new jobs are of the wrong kind or need papers. When there are more technical jobs, the number without work is higher.

8. Do the choices change the answer?

The middle case (50 new jobs per 100 exposed, equal mix, exposed by job title) under different rules:

View	Shortage cut	Gates	Could take no job (%)	Numbers only	Skills, best
strict	0.00	yes	42	50	85
strict	0.00	no	19	50	73
strict	0.30	yes	19	50	73
strict	0.30	no	15	50	62
strict	0.50	yes	4	50	69
strict	0.50	no	4	50	54
broad	0.00	yes	38	50	77
broad	0.00	no	19	50	65
broad	0.30	yes	15	50	69
broad	0.30	no	12	50	58
broad	0.50	yes	0	50	65
broad	0.50	no	0	50	50

- A stricter rule (cut 0) or the strict view raises the count. A looser rule (cut one-half) lowers it. **The result is quite sensitive to the cut**, so real results should show all three.
- Ignoring gates lowers the count by about 10 points. Gates matter.
- If we require the optional tasks too, the share who could take no job rises to 35 in 100.
- If the ropeway operator job needs a certificate, the answer barely changes here.
- Using the other two ways of deciding who is exposed:

9. What we learn for the survey

1. **Keep the three-answer task grid.** Answer 2 changes who could take a job, though not by a lot.
2. **Keep the licence and certificate questions (T3).** Gates decide more than tasks for the driver, guard, guide and technical jobs.
3. **Add direct exposure questions (module D).** Exposure cannot be guessed from job titles or tasks.
4. **Get long task lists for the destination jobs.** The national list is too short for the “unused” measure. A short employer or operator questionnaire can give this, and also the number of jobs.
5. **Report ranges, not one number.** The result depends on the cut, the gate rules, the job supply and the job mix.
6. **More workers in the small groups.** Twenty-six exposed workers give crude results. The real quota should give enough in each of the four exposed jobs.

10. What this cannot tell us

- The job supply is invented. The answer is a function of it.
- Only exposed workers compete for jobs here. In fact others (workers from elsewhere, other locals) will also apply, so real competition is stronger and the count without work is a **lower** figure.
- Workers may also lose part of their work rather than all of it, or move into work that is not on the list (for example their own small business). We do not count that here.
- Having the tasks is not the same as being hired. Age, health, language, caste and contacts also matter and are not in this analysis.
- The task answers were made from our own rules, so agreement with them is not evidence about the world.

11. Files

- `03_analysis/specific_analysis.py`: the code. `03_analysis/results_specific.json`: all numbers.
- `03_analysis/outputs/4_1` to `4_7`: tables and pictures. The destination job list is `4_1_destination_jobs.csv`.